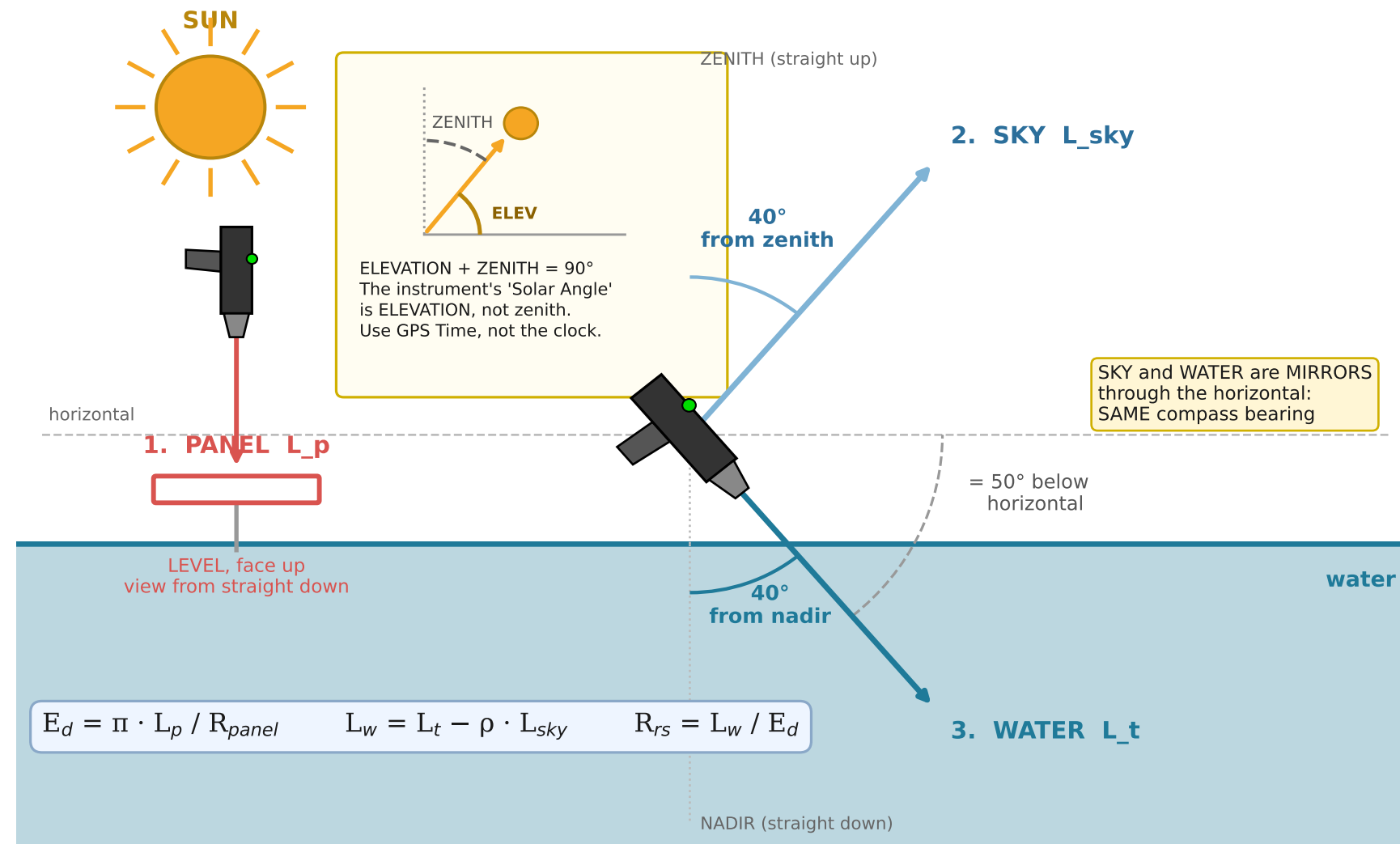


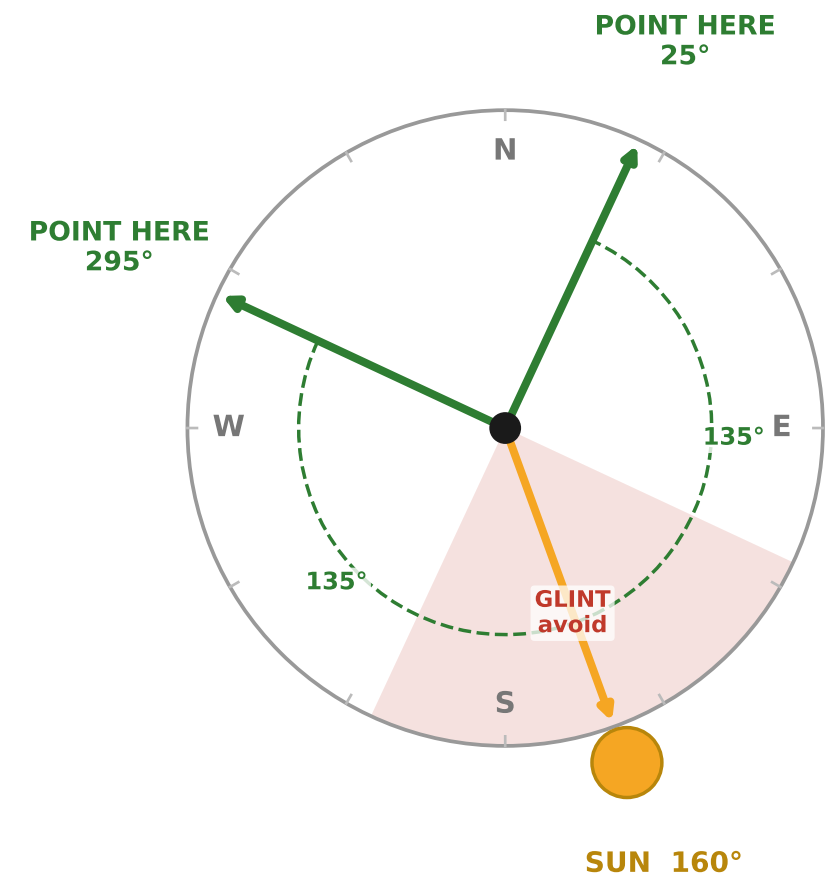
ABOVE-WATER R_{rs} — three-scan field method · Spectral Evolution NaturaSpec Plus

Geometry after Mobley (1999): view 40° from nadir, 135° in azimuth from the sun. Sky scan is the MIRROR of the water scan.

A. SIDE VIEW — the three scans and their angles



B. PLAN VIEW — which way to point (looking down)



Both bearings are 135° in azimuth from the sun. Pick whichever is clear of the boat, its shadow and its wake. The GUI computes these for you from latitude, longitude, date and time.

1. PANEL

White reference panel (Spectralon), LEVEL and face up, in full sun. No shadow on it, yours included. View from straight down. Do not touch the white face.

DARWin: take as the REFERENCE. Re-take every ~10 min, and the moment the light changes.

2. SKY

Sky at 40° from ZENITH = 50° ABOVE horizontal.

SAME compass bearing as the water scan. Not the opposite one. This is the single most common field mistake.

DARWin: TARGET scan. Save.

3. WATER

Water at 40° from NADIR = 50° BELOW horizontal, 135° in azimuth from the sun. FROM A BOAT OR PIER use 90°: 135° looks back at the hull or its shadow (IOCCG v3.0). No glint, foam or wake.

DARWin: TARGET scan. Save. Repeat 2 and 3 five times.

RECORD · WIND

WIND sets ρ , the largest error in the whole measurement. NO ANEMOMETER NEEDED — read it off the water (Beaufort):

- 0-2 < 3 m/s mirror, then wavelets, none breaking p ok
- 3 3.5-5 crests break, SCATTERED whitecaps limit
- 4+ > 5.5 FREQUENT whitecaps INVALID

→ WHITECAPS APPEARING = the edge of where $\rho = 0.028$ holds.

Also: wind DIRECTION, a photo, cloud / sun obscured, panel reflectance (0.99?), and is the water clear or turbid? turbid → do NOT use nir_zero

CHECK ON THE SPOT

⚠ THE CLOCK MAY BE WRONG. Use GPS Time for solar geometry. On a real file the clock read 05:56 with the sun below the horizon.

R_{rs} positive across 400-700? negative = over-subtracted Peak where the water looks? Replicates on top of each other? $R_{rs}(443)$ order 1e-3 to 1e-2?

FOV × range = footprint. An 8° lens at 3.8 m sees ~0.5 m of water; the GUI plots it.

Copy the .sed files off tonight.